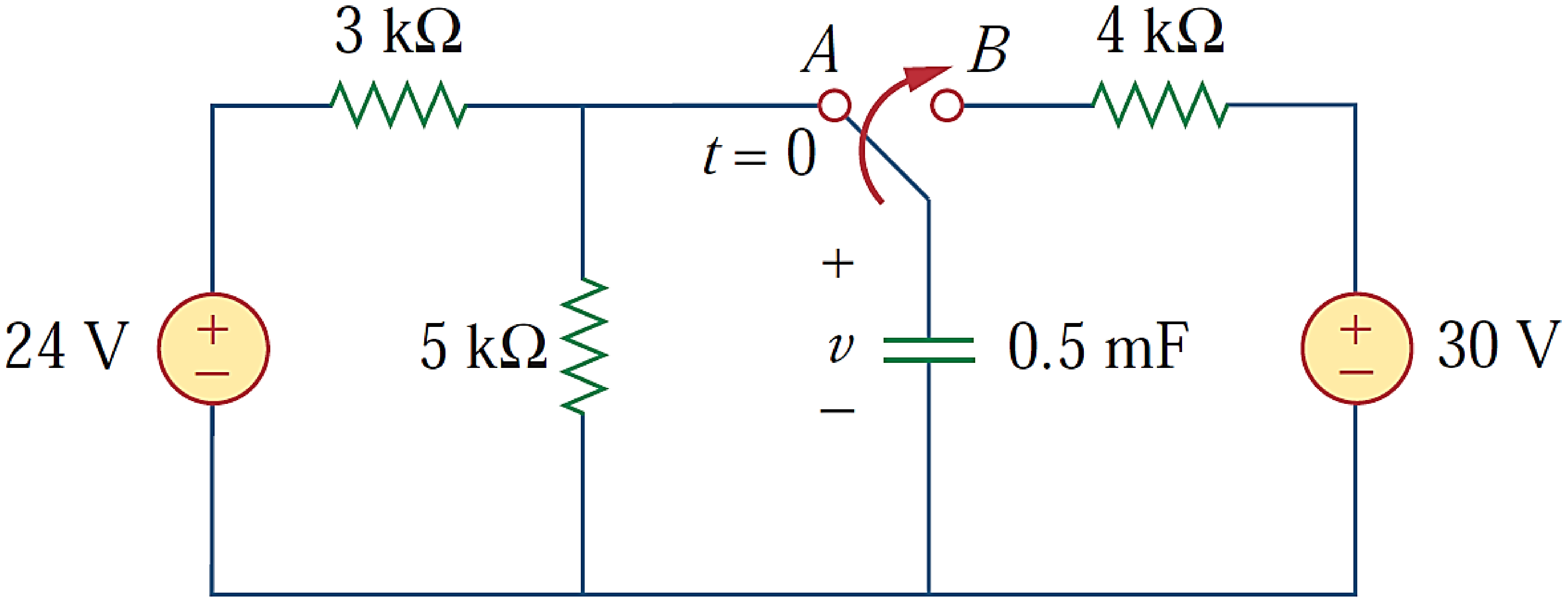
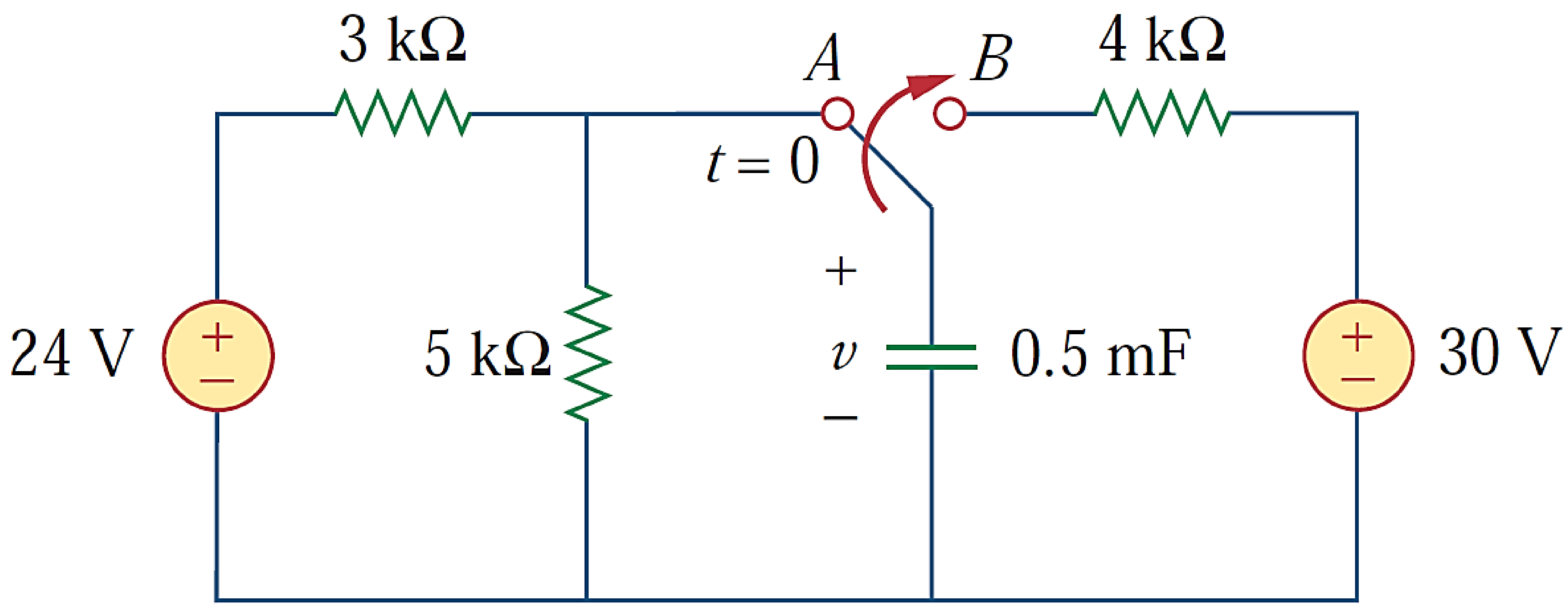
Problem 1: The switch in given circuit was in position A for a long time. At t = *0*, the switch moves to B. Determine the voltage across the capacitor at t = 1s and 4s.





Solution:

For ***t* < 0**, the switch is at position A. The capacitor acts like an open circuit to dc, so *v* is the same as the voltage across the 5kΩ resistor. Hence, the voltage across the capacitor before *t=0* is obtained by voltage division as,

$$\boxed{v(0^-) = \frac{5}{5+3}(24) = 15V}$$

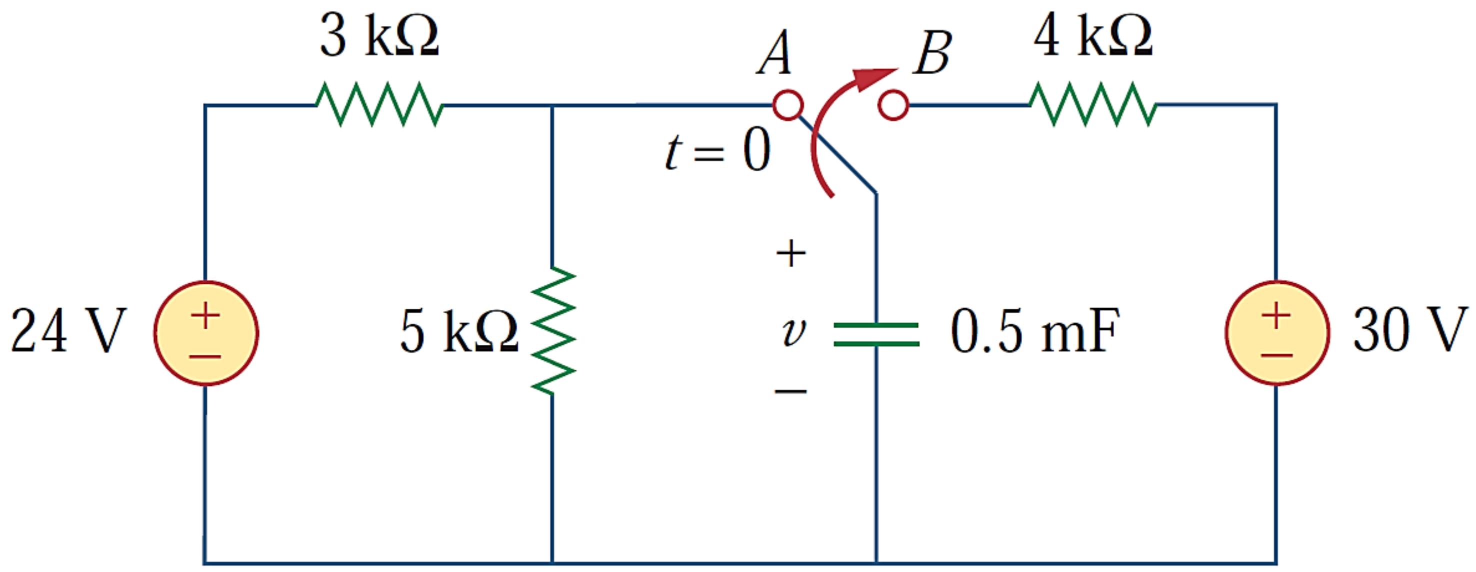
The capacitor voltage cannot change instantaneously,

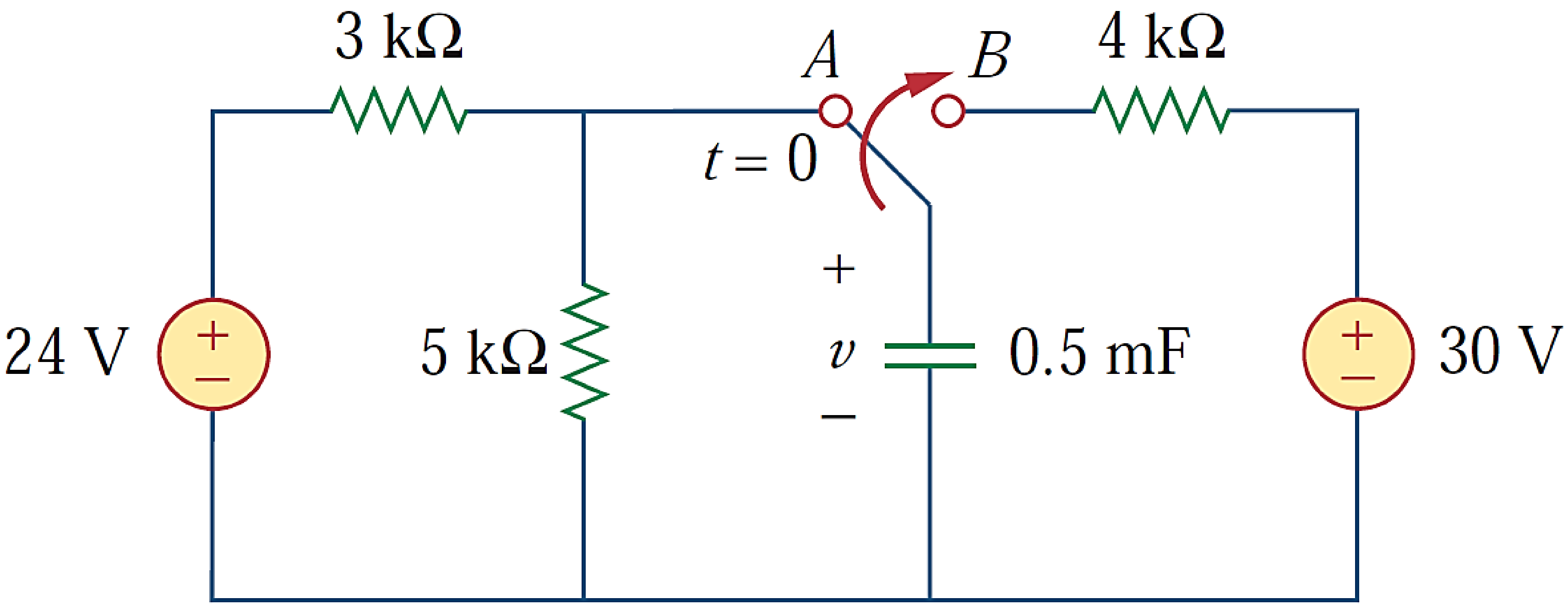
$$\boxed{v(0) = v(0^-) = v(0^+) = 15\text{ V}}$$

For $\boldsymbol{t > 0}$, the switch is in position B. Thevenin resistance connected to the capacitor is $R_{Th} = 4\text{ k}\Omega$, and the time constant is,

$$\boxed{\tau = R_{Th}C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2\text{ s}}$$

$\boxed{v(\infty) = 30\text{ V}}$ (capacitor acts like an open circuit to dc in steady state)





Thus, $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$

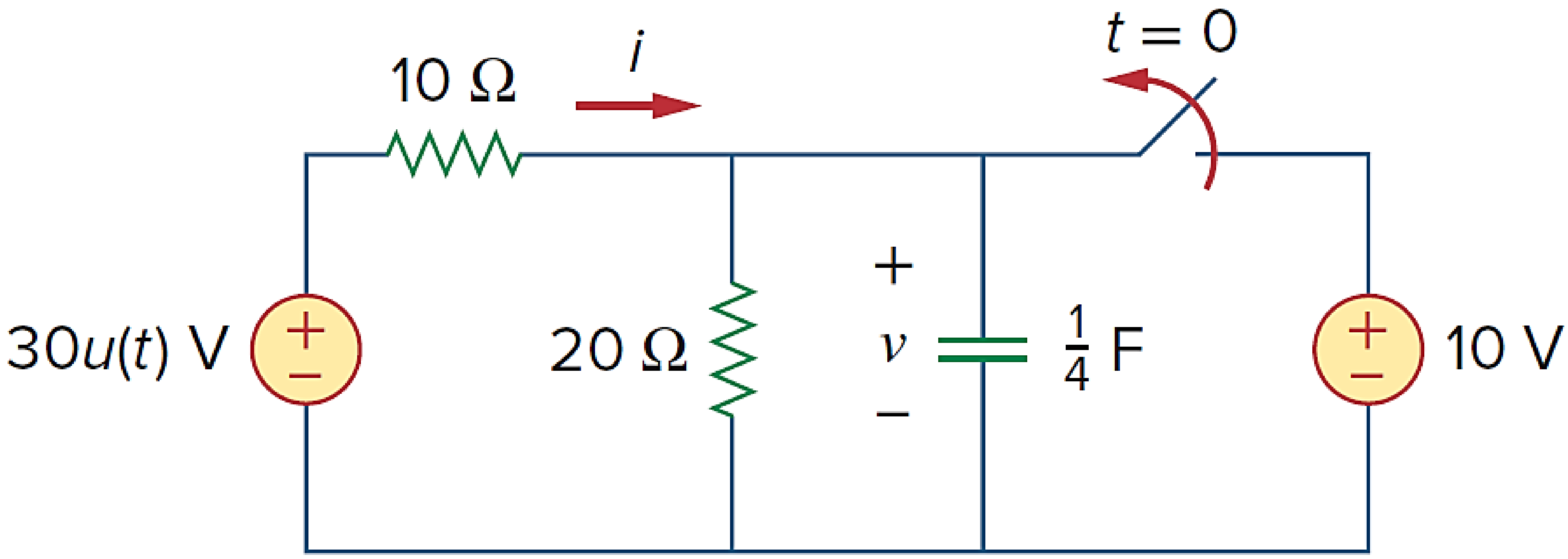
$= 30 + (15 - 30)e^{-t/2}$

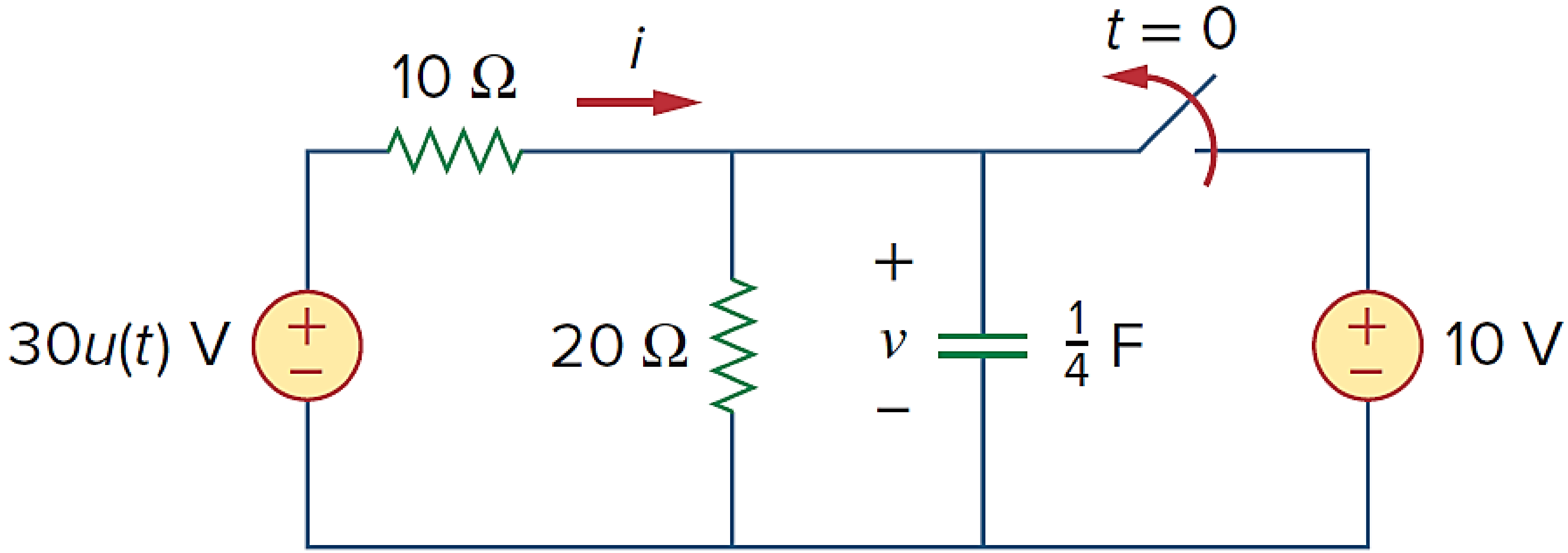
$= (30 - 15e^{-0.5t})$ V

At $t = 1$, $v(1) = 30 - 15\,e^{-0.5} = 20.9$ V

At $t = 4$, $v(4) = 30 - 15e^{-2} = 27.97$ V

Problem 2: In the circuit below, the switch has been closed for a long time and is opened at $t = 0$. Find i and v for all time.



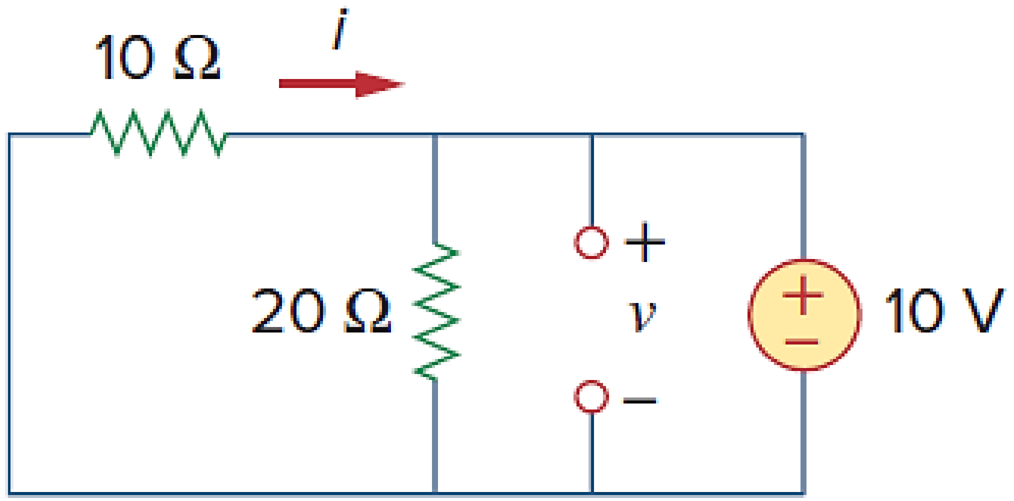


Solution:

Note: The resistor current *i* can be discontinuous at *t=0*, while the capacitor voltage *v* has to be continuous.

Hence, it is always better to find *v* and then obtain *i* from *v*.

For $\boldsymbol{t} < \mathbf{0}$:

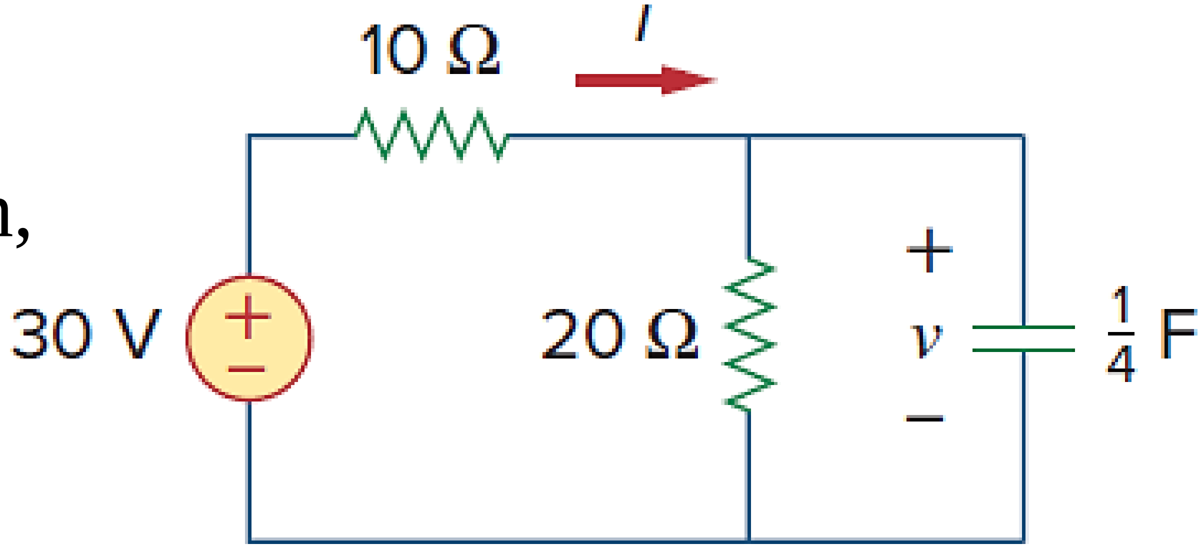


By definition of the unit step function, $30u(t) = \begin{cases} 0 & t < 0 \\ 30 & t > 0 \end{cases}$

For $t < 0$, $\boxed{v = 10V, i = -\dfrac{v}{10} = -1A}$

The capacitor voltage cannot change instantaneously, $\boxed{v(0) = v(0^-) = 10V}$

For $\boldsymbol{t > 0}$, the 10V voltage source is disconnected and the 30V voltage source is now operative



$v(\infty)$ is obtained using voltage division,

$$\boxed{v(\infty) = \frac{20}{20+10}(30) = 20V}$$

The Thevenin resistance at the capacitor terminals is $\boxed{R_{Th} = 10 \| 20 = 20/3\ \Omega}$

Time constant is $\boxed{\tau = R_{Th}C = \frac{20}{3}.\frac{1}{4} = \frac{5}{3}s}$

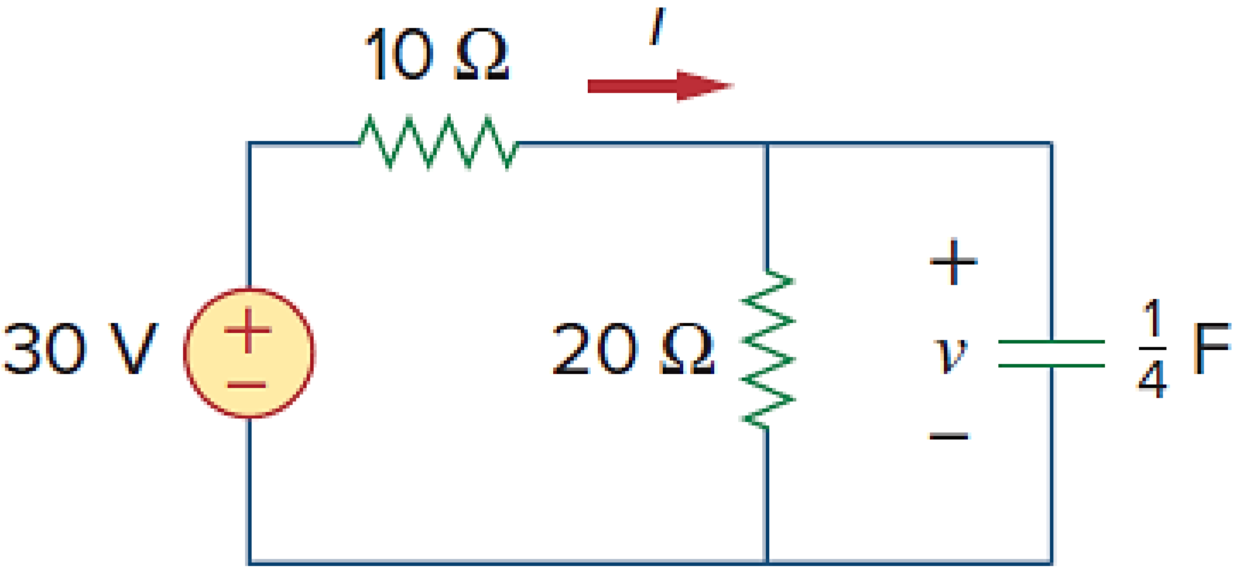
Thus, $$\boxed{\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)\ ]e^{-t/\tau} \\ &= 20 + (10 - 20)e^{-(3/5)t} = (20 - 10e^{-0.6t})\ \text{V} \end{aligned}}$$

i is the sum of the currents through the 20-Ω resistor and the capacitor,

$$\boxed{i = \frac{v}{20} + C\frac{dv}{dt}}$$

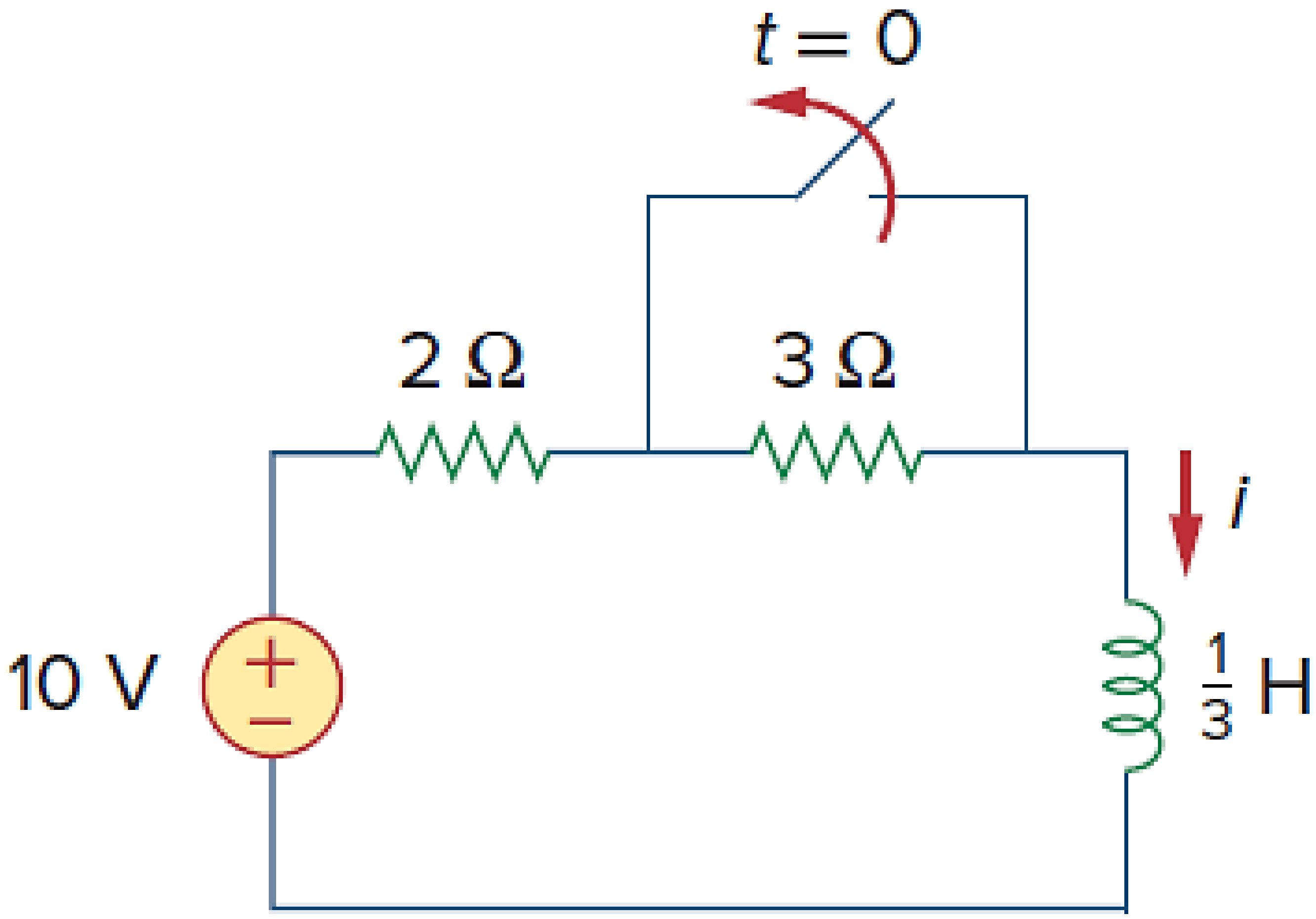
$$i = 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} = (1 + e^{-0.6t})\ \text{A}$$

Or $\boxed{v + 10i = 30}$ (KVL in outer loop)



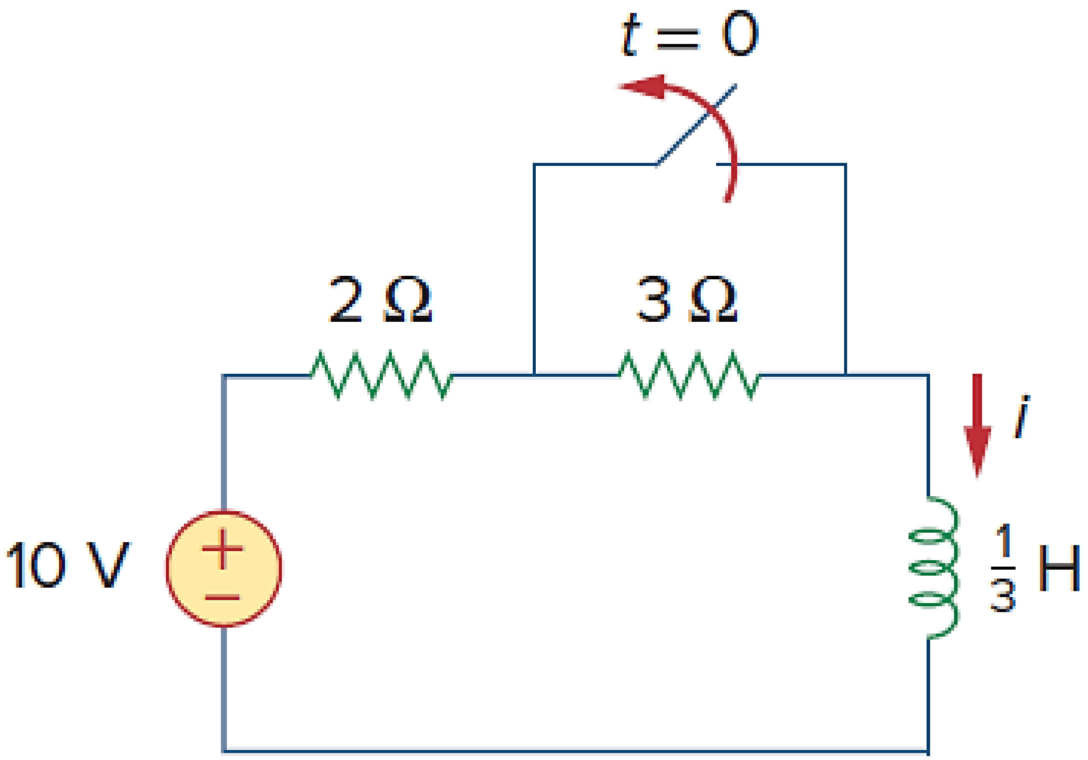
$$\boxed{\begin{aligned} v &= \begin{cases} 10V & t < 0 \\ (20 - 10e^{-0.6t})V & t > 0 \end{cases} \\ i &= \begin{cases} -1A & t < 0 \\ (1 + e^{-0.6t})\ \text{A} & t > 0 \end{cases} \end{aligned}}$$

Problem 3: Find *i*(t) in the circuit below for *t* > 0. Assume that the switch has been closed for a long time before opening.



Solution:

For $\boldsymbol{t < 0}$, the 3-Ω resistor is short-circuited, and the inductor acts like a short circuit. The current through the inductor at $t = 0^-$ is $\boxed{i(0^-) = 10/2 = 5\text{A}}$



Since inductor current cannot change instantaneously $\boxed{i(0) = i(0^+) = i(0^-) = 5\text{A}}$

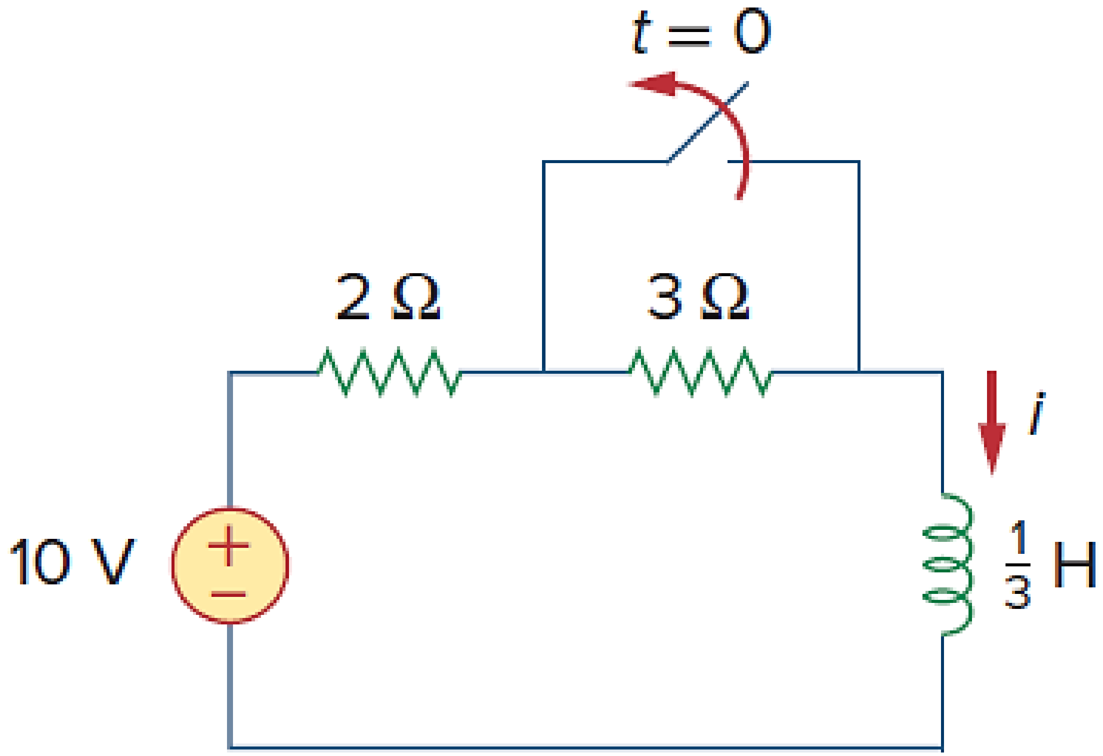
For $\boldsymbol{t > 0}$, the 3-Ω resistor comes in series with the 2-Ω resistor

The steady-state inductor current is $\boxed{i(\infty) = \dfrac{10}{2+3} = 2\text{A}}$

The Thevenin resistance across the inductor terminals is

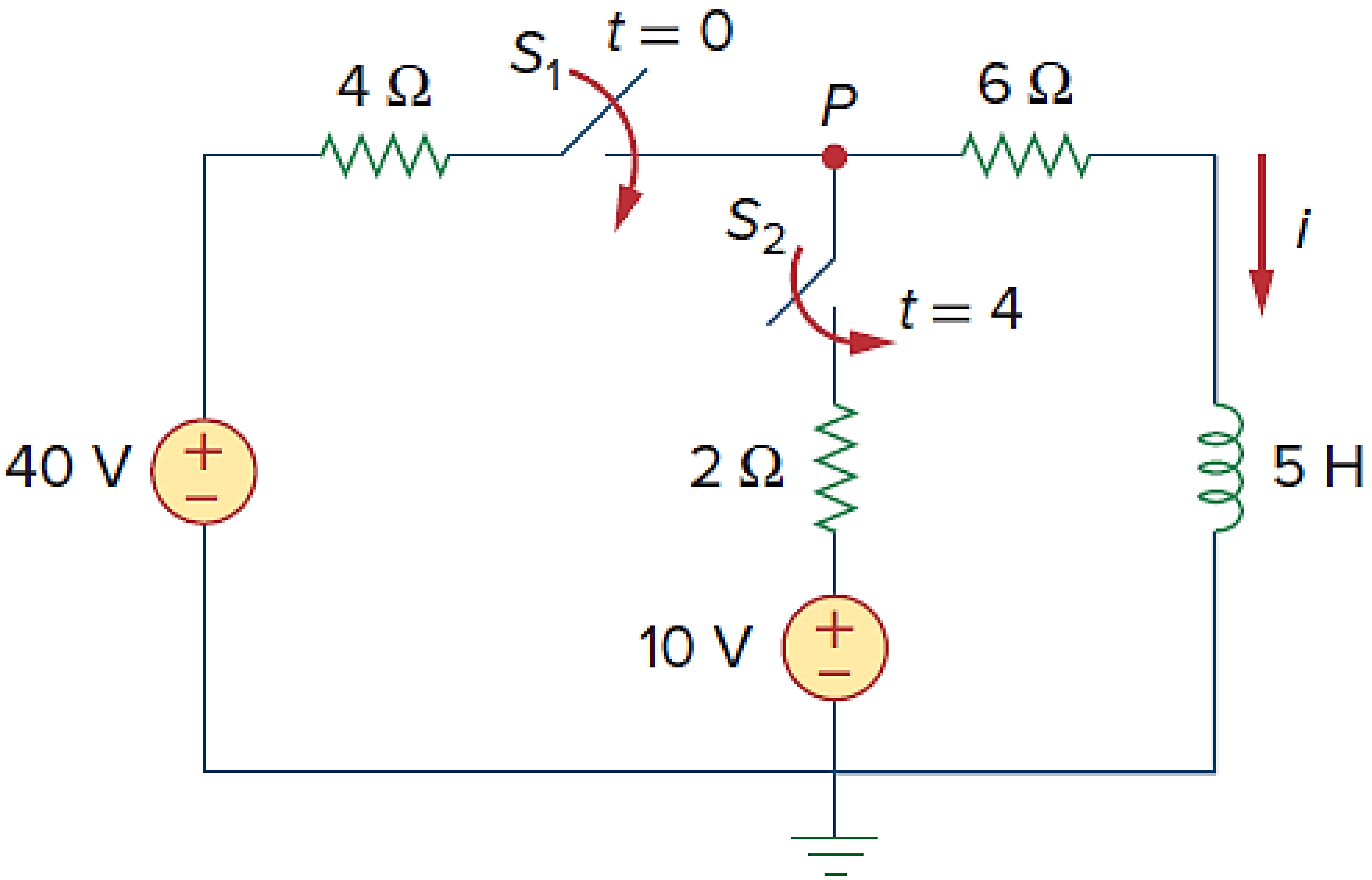
$$\boxed{R_{TH} = 2 + 3 = 5\ \Omega}$$

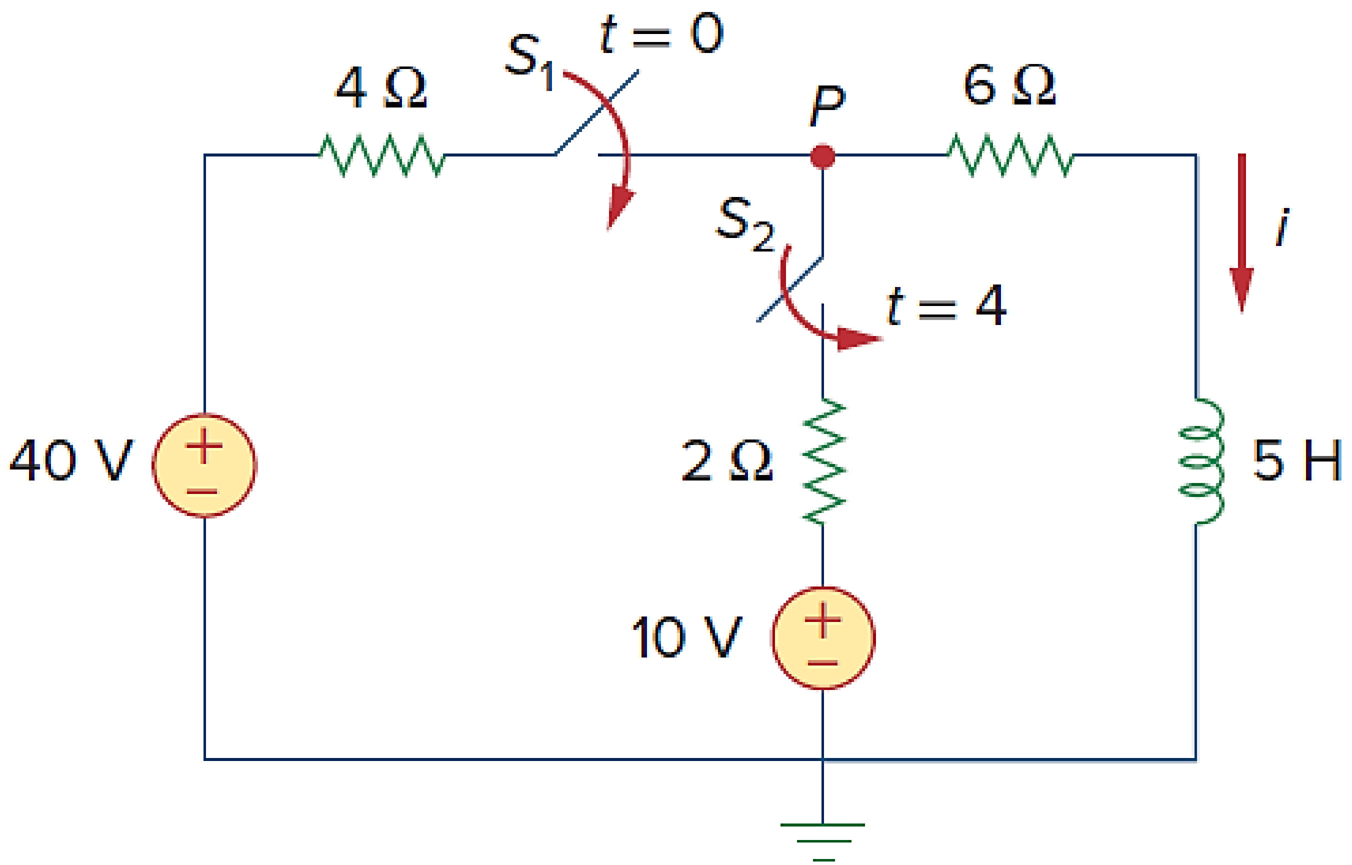
Time constant $\boxed{\tau = \frac{L}{R_{TH}} = \frac{\frac{1}{3}}{5} = \frac{1}{15}\ s}$



$$\boxed{\begin{aligned} i(\mathrm{t}) &= i(\infty) + [i(0) - i(\infty)]e^{-t/\tau} \\ &= 2 + (5 - 2)e^{-15t} \\ &= 2 + 3e^{-15t}\ \mathrm{A}, \quad t > 0 \end{aligned}}$$

Problem 4: At $t = 0$, switch 1 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for $t > 0$. Calculate i at $t = 2$ s and $t = 5$ s.



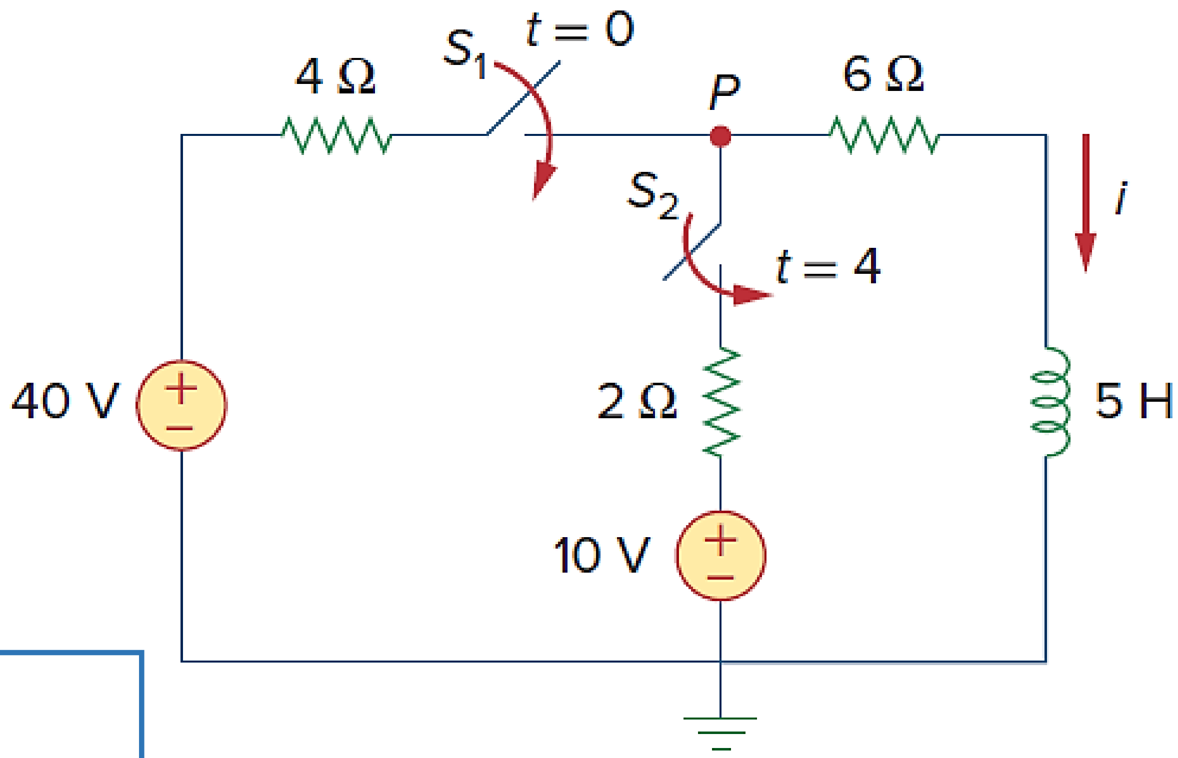


Solution:

Solve for the three time intervals $t \le 0$, $0 \le t \le 4$, and $t \ge 4$ separately.

For $\boldsymbol{t} < \mathbf{0}$, switches S_1 and S_2 are open so that $i = 0$. Since the inductor current cannot change instantly, $\boxed{i(0) = i(0^+) = i(0^-) = 0\text{A}}$

For $\mathbf{0 \le t \le 4}$, S_1 is closed so that the 4-Ω and 6-Ω resistors are in series.



$$\boxed{i(\infty) = \frac{40}{4+6} = 4A;\ R_{Th} = \ 4+6 = 10\Omega}$$

$$\boxed{\tau = \frac{L}{R_{TH}} = \frac{5}{10} = \frac{1}{2}s}$$

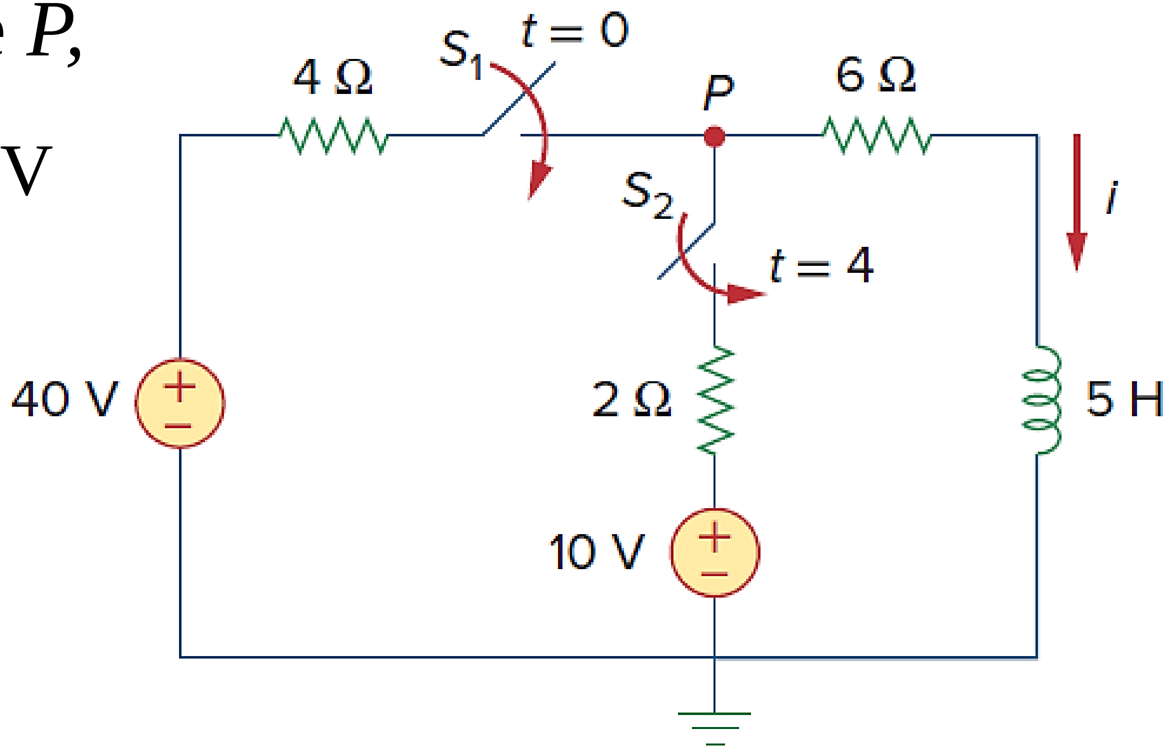
Thus
$$\boxed{\begin{aligned} i(t) &= i(\infty) + [i(0) - i(\infty)]e^{-t/\tau} \\ &= 4 + (0-4)e^{-2t} = 4(1 - e^{-2t})\ \text{A} \end{aligned}}$$

For $\mathbf{t \ge 4}$, S_1 and S_2 are closed; the 10-V voltage source is also connected into the circuit. The initial current is, $\boxed{i(4) = 4(1 - e^{-8}) \simeq 4\ \text{A}}$

To find $i(\infty)$, let v be the voltage at node P,

Using KCL, $\dfrac{40-v}{4}+\dfrac{10-v}{2}=\dfrac{v}{6}$; $v=\dfrac{180}{11}$ V

$$\boxed{i(\infty)=\frac{v}{6}=\frac{30}{11}=2.72\text{A}}$$



The Thevenin resistance at the inductor terminals is $\boxed{R_{Th} = 4\|2 + 6 = (22/3)\Omega}$

Time constant

$$\boxed{\tau=\frac{L}{R_{TH}}=\frac{5}{\frac{22}{3}}=\frac{15}{22}s}$$

$$\boxed{\begin{aligned} i(t) &= i(\infty) + [i(4) - i(\infty)]e^{-(t-4)/\tau}, \quad \text{for } \boldsymbol{t \geq 4} \\ &= 2.727 + (4 - 2.727)e^{-(t-4)/\tau} \quad (\tau = 15/22) \\ &= 2.727 + 1.273e^{-1.4667(t-4)} \end{aligned}}$$

Putting everything together,

$$\boxed{i(t) = \begin{cases} 0 & t \le 0 \\ 4(1 - e^{-2t}) & 0 \le t \le 4 \\ 2.727 + 1.273e^{-1.4667(t-4)} & t \ge 4 \end{cases}}$$

At $\boldsymbol{t}$ **= 2s,** $\boldsymbol{i}\mathbf{(2)} = 4(1 - e^{-4}) = \mathbf{3.93\ A}$

At $\boldsymbol{t}$ **= 5s,** $\boldsymbol{i}\mathbf{(5)} = 2.727 + 1.273e^{-1.4667} = \mathbf{3.02\ A}$